Experiment # 09

Implementation of Image Segmentation

Objective

1. To implement image segmentation

Software

- Python Idle 3.7 or 3.8.5
- Pycharm

Theory

Segmentation of an image can be achieved by clustering: a technique that groups together the pixels in an image using some measure of similarity.

K means clustering is the most basic and widely used clustering technique for data grouping, in machine learning and in colored image segmentation.

The algorithm for K-means clustering is as following

1. Chose the number (K) of clusters and randomly select the centroids of each cluster.
2. For each data point:
   - Calculate the distance from the data point to each cluster.
   - Assign the data point to the closest cluster.
3. Recompute the centroid of each cluster.
4. Repeat steps 2 and 3 until there is no further change in the assignment of data points (or in the centroids).

Thresholding is the simplest method of image segmentation. From a grayscale image, thresholding can be used to create binary images. For example in the image shown below threshold is applied around the gray value 160 and we have easily extracted the hand out of the

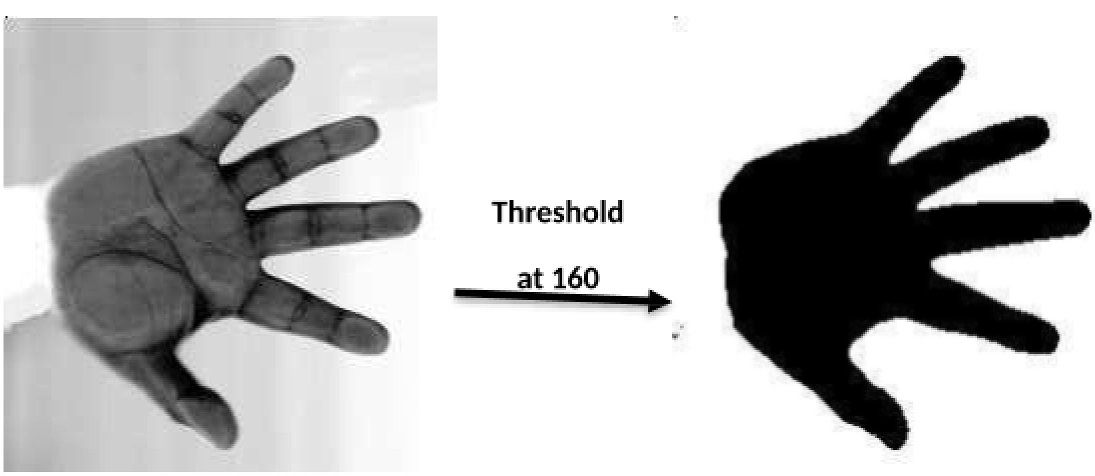
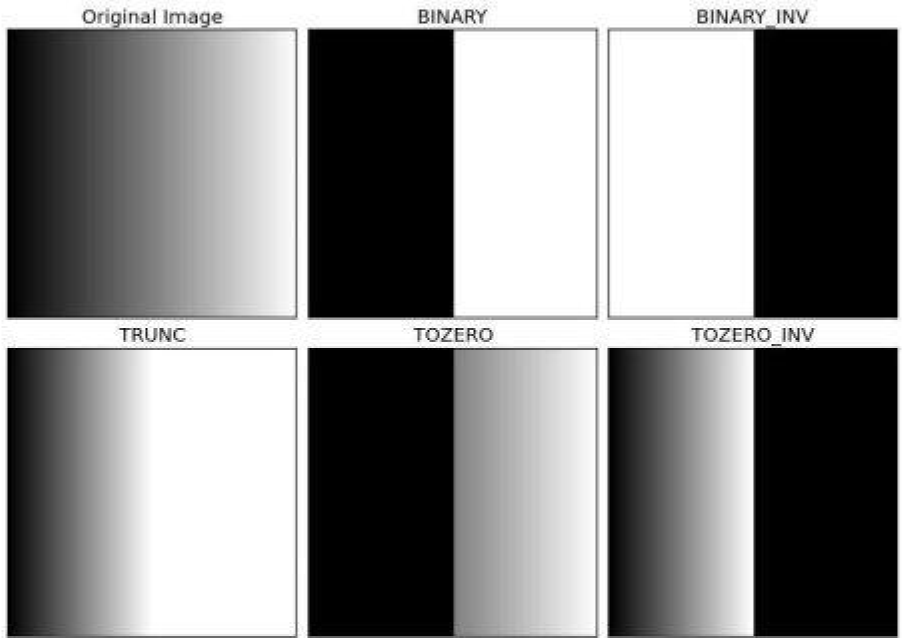


image.



Adaptive thresholding: In some cases a simple threshold doesn't work. An alternative approach is local threshold which is to statistically examine the intensity values of the local neighborhood of each pixel and threshold on the bases of local mean, median or mode etc.

Some Useful Commands:

1. OpenCV reads image in BGR format by default
   Cv2.kmeans (image , k , label , criteria , attempt , flags)

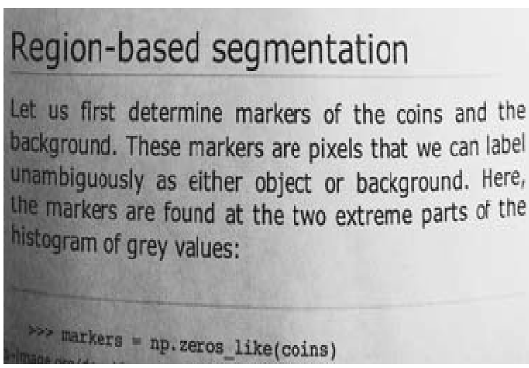
   Where criteria = (cv2.TERM_CRITERIA_EPS+cv2.TERM_CRITERIA_MAX_ITER, 100, 0.2)
2. cv2.threshold(img,127,255,cv2.THRESH_BINARY)
3. cv2.adaptiveThreshold(src, dst, maxValue, adaptiveMethod, thresholdType, blockSize, C)

- Adaptive method:
   1. ADAPTIVE_THRESH_MEAN_C − threshold value is the mean of neighborhood area.
   2. ADAPTIVE_THRESH_GAUSSIAN_C − threshold value is the weighted sum of neighborhood values where weights are a Gaussian window.

Lab Task:

1. Apply K-mean clustering algorithm on given image for color quantization where K =3, 8, 13, 20 and 40.

2. Apply thresholding (binary and adaptive) on the given images such that all letters extracted out from the image



Region-based segmentation

Let us first determine markers of the coins and the background. These markers are pixels that we can label unambiguously as either object or background. Here, the markers are found at the two extreme parts of the histogram of grey values:

```
>>> markers = np.zeros_like(coins)
```